

GSP

Hyperparameters in GSP:

Epochs	30
Batch size	128
Learning rate	0.05
Momentum	0.9
Weight decay	1e-3
Number of groups	4
Geometric temperature	0.5
Geometric loss weight	1e-2
Orthogonality loss weight	1e-3
Pruning cycles	5
Groups	4
Target prune rate	0.5
Fusion cosine threshold	0.95

Hyperparameters that need to be tuned:

1)cifar_gdsp.py:

- **--tau-geo** — controls how “soft” vs “peaky” assignments are; strongly affects clustering stability.
- **--lambda-geo** and **--lambda-ortho** — balance task loss vs alignment vs centroid diversity; main knobs for accuracy vs grouping behavior.
- **-g / --groups** — capacity of the grouping; affects redundancy and downstream pruning.
- **--lr**, **--epochs**, optionally **--batch-size** — standard training; G-DSP adds extra terms so LR may need adjustment.

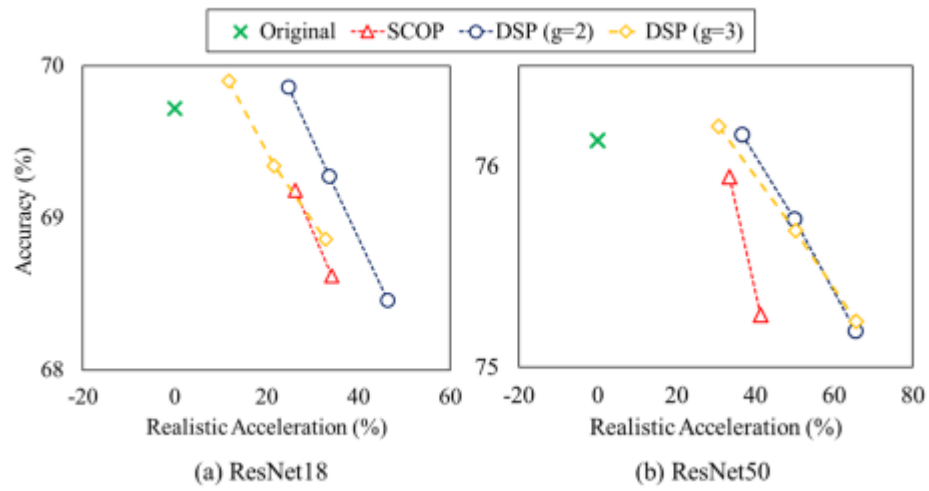
2)cifar_finetrain.py:

- **-p / --prune** — target compression.
- **-c / --cycles** and **--epochs / --lr** — stability after repeated pruning.
- **--merge-threshold** — trade-off between more fusion (smaller effective model) vs accuracy.

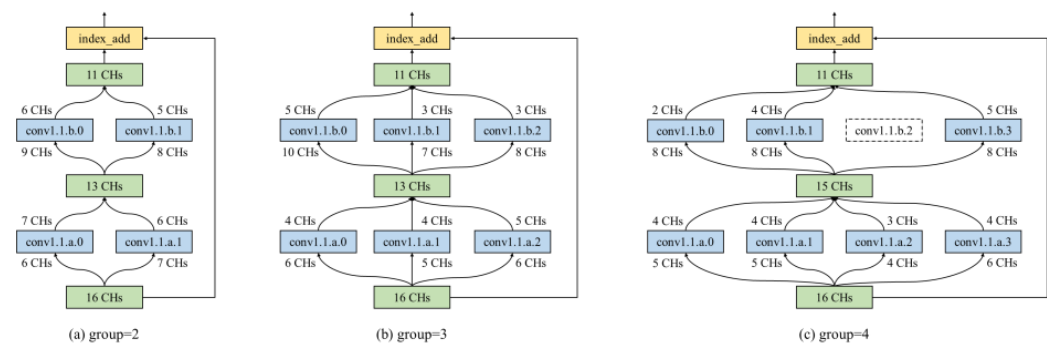
Parameters that need to be noted/compared:

- 1) P. Acc. (%) , Δ Acc. (%) , Params $\downarrow$ (%) ,FLOPs $\downarrow$ (%)
- 2) Top-1 P. Acc. (%) Top-1 Δ Acc. (%) Top-5 P. Acc. (%) Top-5 Δ Acc.

Plotting:



1)



2)